

RelayGuard ML System Documentation

(Engineer Guide)

RelayGuard Team

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1 System Overview

Sensor Data $\rightarrow$ Physics Model (HI) $\rightarrow$ ML Model $\rightarrow$ RUL

Explanation:

The system does **not** directly predict failure from raw sensor data. Instead:

1. Sensor data is converted into a **Health Index (HI)** using physics models.
2. The ML model learns patterns from HI and engineered features.
3. The final output is the **Remaining Useful Life (RUL)**.

2 Input Features (What We Measure)

Parameter	Symbol	Meaning	Used In
Current	I	Load through relay	Arc damage model
Voltage	V	Switching voltage	Arc damage model
Temperature	T	Relay heat	Thermal model
Cycles	N	Number of operations	Mechanical wear
Resistance	R	Contact degradation	Direct wear indicator
Temp Rate	dT/dt	Heating speed	ML anomaly detection
Peak Current	I_{peak}	Current spikes	ML stress feature
Frequency	f	Switching rate	ML usage intensity

3 Physics-Based Modeling (Core Logic)

3.1 Arc Damage Model

$$E = I^2 \cdot V \cdot t$$

Why used:

- Models **contact erosion during switching**.

- Damage increases rapidly with current.

$$W_{total} = \sum (E \cdot K_T)$$

Meaning:

- Total accumulated damage over the relay's lifetime.
- Used as a long-term degradation indicator.

3.2 Thermal Model (Arrhenius)

$$K_T = e^{\left(\frac{E_a}{k} \left(\frac{1}{T_{ref}} - \frac{1}{T}\right)\right)}$$

Why used:

- Temperature accelerates aging of contact materials.
- High temperature = faster failure rate.

3.3 Resistance Model

$$R = R_0 + kN^\alpha$$

Why used:

- Contact resistance increases with each switching cycle.
- Provides a direct, measurable degradation signal.

3.4 Composite Health Index (HI)

$$HI = 0.40 \cdot HI_{damage} + 0.25 \cdot HI_{thermal} + 0.25 \cdot HI_{resistance} + 0.10 \cdot HI_{cycles}$$

Why used:

- Combines all degradation mechanisms into a single value.
- Gives a **single health score (0–100%)** for easy monitoring.

Parameter	Used In	Purpose
I	Damage model	Arc energy
V	Damage model	Arc intensity
T	Thermal model	Aging acceleration
N	Cycle model	Mechanical wear
R	Resistance model	Direct degradation
dT/dt	ML only	Detect abnormal heating
I_{peak}	ML only	Capture spikes
f	ML only	Usage rate

4 Where Each Parameter is Used

5 ML Training Strategy

5.1 Key Concept

We do **NOT** train on raw sensor data directly. Instead:

Processed Features $\rightarrow$ Future HI or RUL

Why:

- The physics model provides **structured, meaningful data**.
- ML focuses on learning **patterns over time**.
- This approach improves accuracy and generalization.

6 Feature Engineering

Final features used for ML training:

- Current (I)
- Voltage (V)
- Temperature (T)
- Cycles (N)

- Resistance (R)
- Temperature rate (dT/dt)
- Peak current (I_{peak})
- Frequency (f)
- Cumulative damage (W_{total})

Purpose: Capture both **instant stress** (peaks, rates) and **long-term degradation** (cumulative damage, resistance).

7 Dataset Structure

```
features = [
    "current", "voltage", "temp",
    "cycles", "resistance",
    "dT_dt", "peak_current", "frequency",
    "damage"
]

target = "HI_future"
```

Explanation:

- **Input** = current system state (sensor readings + derived features).
- **Output** = future degradation (HI at next time step).
- Enables prediction **before** failure actually occurs.

8 Model Training

8.1 Option 1: XGBoost (Recommended for Round 1)

```
from xgboost import XGBRegressor

model = XGBRegressor(
    n_estimators=100,
    max_depth=5,
    learning_rate=0.1
)

model.fit(X_train, y_train)
```

Why XGBoost:

- Works well with tabular/structured data.
- Fast to train and highly accurate.
- Does not require a huge dataset to perform well.

8.2 Option 2: LSTM (Advanced)

- **Input:** time-series window (e.g., last 50 readings)
- **Output:** future HI or RUL

Why LSTM:

- Captures temporal patterns and trends in sensor data.
- Better suited for long-term RUL prediction.

9 Prediction Pipeline

Sensor Data $\rightarrow$ HI $\rightarrow$ ML Model $\rightarrow$ RUL

Steps:

1. Read live sensor data.
2. Compute Health Index (HI) using physics models.
3. Generate engineered features from HI and sensors.
4. Feed features into the ML model to predict future HI/RUL.

10 Alert System

HI (%)	Status
> 70	Healthy
40–70	Warning
20–40	Critical
< 20	Failure Imminent

Usage:

- Drives real-time dashboard alerts for operators.
- Enables proactive maintenance scheduling.
- Can trigger automatic relay switching logic.

11 Full System Pipeline

1. **Sensors** — collect current, voltage, temperature, resistance, cycles
2. **Physics Model (HI)** — compute arc damage, thermal aging, resistance wear
3. **Feature Engineering** — combine raw + derived features
4. **ML Model** — learn degradation patterns
5. **Prediction (RUL)** — estimate remaining useful life
6. **Alerts** — trigger warnings, maintenance actions

12 Key Insight

Physics Model = Ground Truth

ML Model = Pattern Learning

Interpretation:

- Physics ensures **correctness** and domain knowledge.
- ML improves **prediction accuracy** from historical patterns.
- Together → reliable, explainable predictive maintenance.

13 Final Understanding

- Physics converts raw signals into **meaningful degradation indicators**.
- ML learns how degradation **evolves over time**.
- The combined system predicts relay failure **before it happens**, enabling proactive maintenance and preventing unplanned downtime.